MHT-CET Weightage: 3-4%

Viscosity & Flow Regimes

Viscosity Laws

Newton's law of viscosity

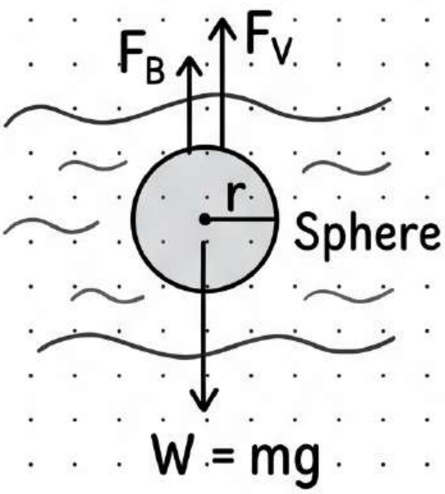
$$F = \eta A \frac{\Delta V_x}{\Delta y}$$

SI Units of η : $\frac{N \times s}{m^2}$

Poiseuille's formula

$$Q = \frac{dV}{dt} = \frac{\pi p r^4}{8 \eta L}$$

Stoke's Law & Terminal Velocity



Stoke's Law

Viscous force $F_V = 6 \pi \eta r v$

Terminal velocity

$$V_T = \frac{2}{9} \frac{r^2 (\rho - \sigma) g}{\eta}$$

Therefore, $V_T \propto r^2$

Reynolds Number (R_e)

$$R_e = \frac{\rho V d}{\eta}$$

- ☐ If $R_e < 1000$, then the flow is laminar.
- ☐ If $R_e > 2000$, then the flow is turbulent.

Surface Tension & Capillarity

Surface Tension & Energy

Surface Tension (T) = $\dfrac{\text{Total force}}{\text{Length of line } (\ell)}$

Surface Energy (U) = Surface Tension (T) × Exposed Area (A)

For liquid drop or water bubble $A = 4\pi r^2$, and for soap bubble $A = 8\pi r^2$

Splitting of Drops

Splitting of bigger drop into smaller droplets: $R = n^{1/3} r$ (R = radius bigger drop, r = radius smaller drops)

Work done = Change in surface energy = $4\pi R^2 T(n^{1/3} - 1)$

Excess Pressure ($P_{ex} = P_{in} - P_{out}$)

In liquid drop: $P_{ex} = \dfrac{2T}{R}$

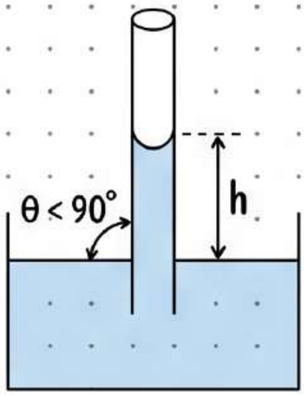
In soap bubble: $P_{ex} = \dfrac{4T}{R}$

Angle of Contact & Capillarity

If θ < 90° : Concave shape, Liquid rise up in capillary

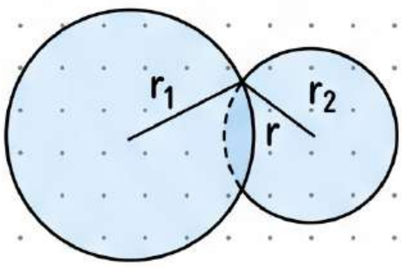
If θ > 90° : Convex shape, Liquid falls down in capillary

Capillary rise $h = \dfrac{2T\cos\theta}{r\rho g}$



When two soap bubbles are in contact, radius of curvature of common surface is

$$r = \frac{r_1 r_2}{r_1 - r_2} \quad (r_1 > r_2)$$



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Fluid Dynamics & Bernoulli's

Types of Flow

- Steady: Fluid characteristics at a point do not change with time.
- Streamline: All particles passing through a given point follow the same path.
- Laminar: Fluid particles move along well-defined streamlines which are straight and parallel.
- Turbulent: Motion of fluid particles is random.

Fundamental Equations

Equation of continuity:

$$A_1v_1 = A_2v_2$$

(Based on conservation of mass)

Bernoulli's theorem:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

(Based on the conservation of energy)

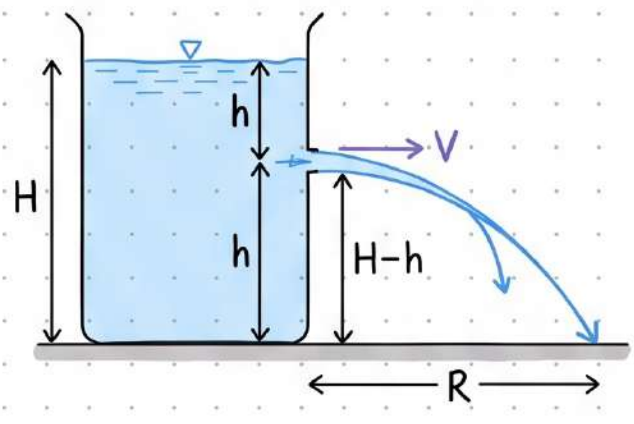
Energies (per unit volume)

Kinetic energy per unit volume = $\frac{1}{2}\rho v^2$.

Potential energy per unit volume = ρgh.

Pressure energy per unit volume = P.

Efflux & Flow Rate



Volume of water flowing per second

$Q = A_1v_1 = A_2v_2$.

Velocity of efflux

$V = \sqrt{2gh}$.

Horizontal range $R = 2\sqrt{h(H-h)}$.

Fluids: Statics & Buoyancy

Properties of Fluid

Density = $\frac{\text{mass}}{\text{volume}}$ (kgm^{-3})

Specific weight = $\frac{\text{weight}}{\text{volume}} = \rho g$ $(\text{kg m}^{-2}\text{s}^{-2})$

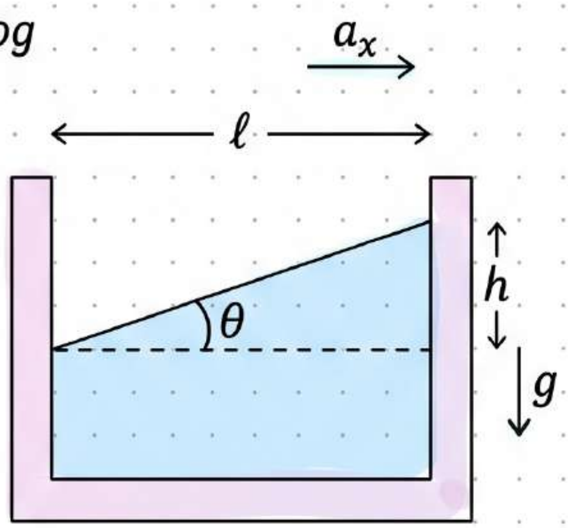
Relative density = $\frac{\text{density of given liquid}}{\text{density of pure water at 4°C}}$ (Unitless)

Pressure = $\frac{\text{normal force}}{\text{area}} = \frac{\text{thrust}}{\text{Area}}$ (Nm^{-2})

Variation of Pressure

Variation of pressure with depth h: $\Delta P = h\rho g$

Pressure in Accelerating Fluid



Liquid placed in elevator: When elevator accelerates upward with acceleration a_0 then pressure in the fluid at depth h is given $P = h\rho[g + a_0]$.

Container accelerating in horizontal and vertical directions:

$$\tan\theta = \frac{h}{\ell} = \frac{a_x}{g + a_y} \quad \text{(going up, i.e., } a_y = +\text{)}$$

For container going down ($a_y = -$): $\tan\theta = \frac{h}{\ell} = \frac{a_x}{g - a_y}$

Buoyancy

Buoyant force = weight of displaced liquid = $\rho_l \cdot v_l \cdot g$
Here ρ_l = density of liquid, v_l = volume of displaced liquid.